

Wandering Black Hole Feeds on Its Gravitational Wake

A 35,000-solar-mass black hole accretes from a dense downstream wake, revealing a new growth channel.

arXiv:2608.10719 — Multiwavelength evidence for off-nuclear intermediate-mass black hole accretion.

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Background

Intermediate-mass black holes are thought to seed supermassive black holes, but many wander far from galactic nuclei. Their fuel supply has remained uncertain because normal fueling channels are inaccessible.

Direct observational evidence for gravitational-wake accretion onto wandering black holes has been elusive.

What this paper is selling

This paper reports a 35,000-solar-mass wandering black hole accreting through a gravitational wake. Multi-epoch spectroscopy reveals upstream flow, downstream wake, and optically thick absorbers driving rapid broad-line variability.

Key results to remember

- Broad lines, compact continuum, variability, and power-law SED confirm black hole nature.
- Three gas components: blueshifted upstream flow, redshifted dense downstream wake, optically thick absorbers.
- Absorbers within capture radius cause rapidly changing-look broad-line emission.

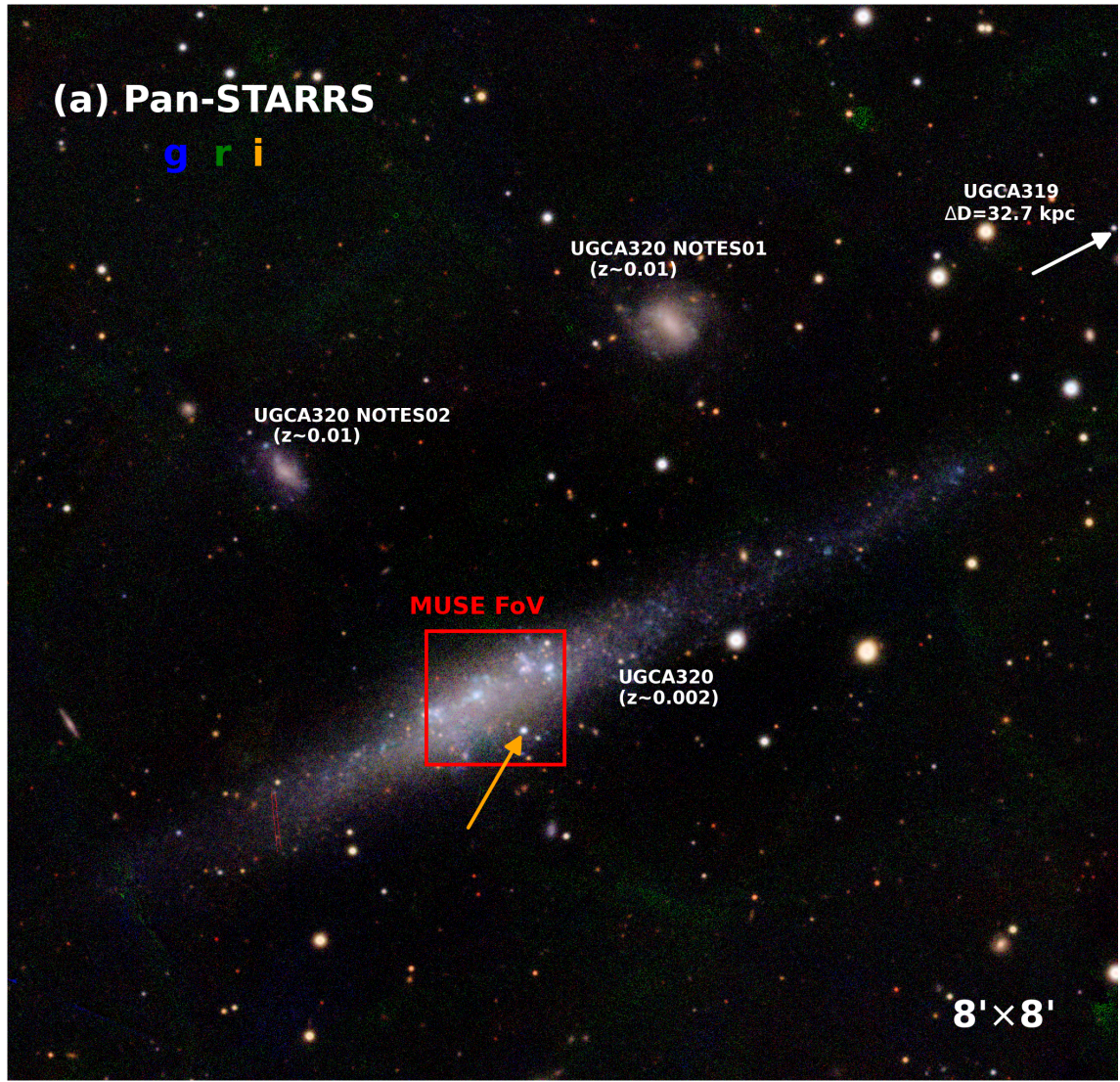


Figure 1 | HST and MUSE views pinpoint the off-nuclear IMBH in UGCA 320, with velocity map revealing disturbed gas.

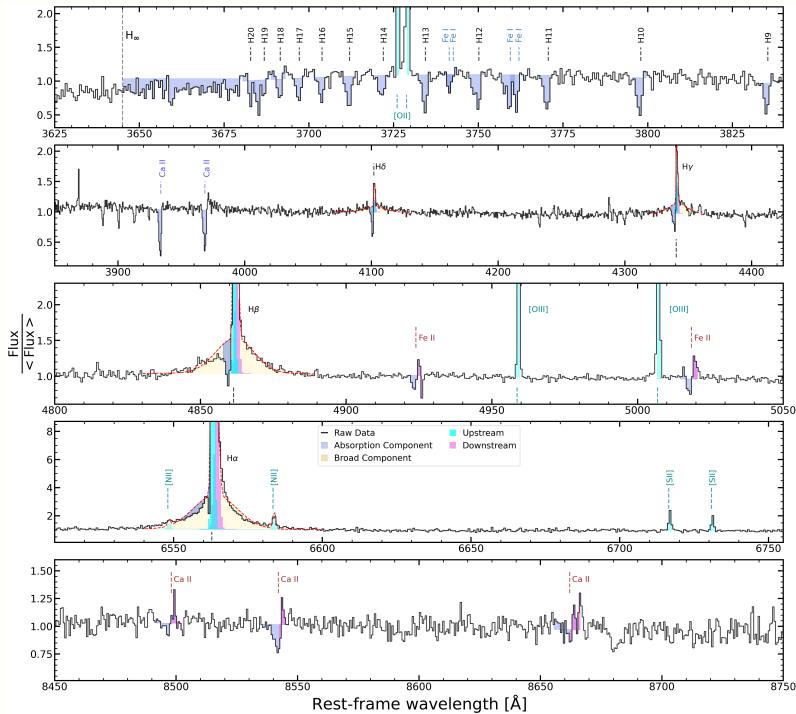


Figure 3 | X-Shooter spectrum decomposes blueshifted absorption, upstream and downstream emission, and broad AGN lines.